



By Nitesh Singh

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SOC

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Lateral Movement

What is Lateral Movement?

Lateral Movement is when an attacker moves from one compromised system to other systems inside a network.

Instead of leaving the network, the attacker **moves sideways across machines**.

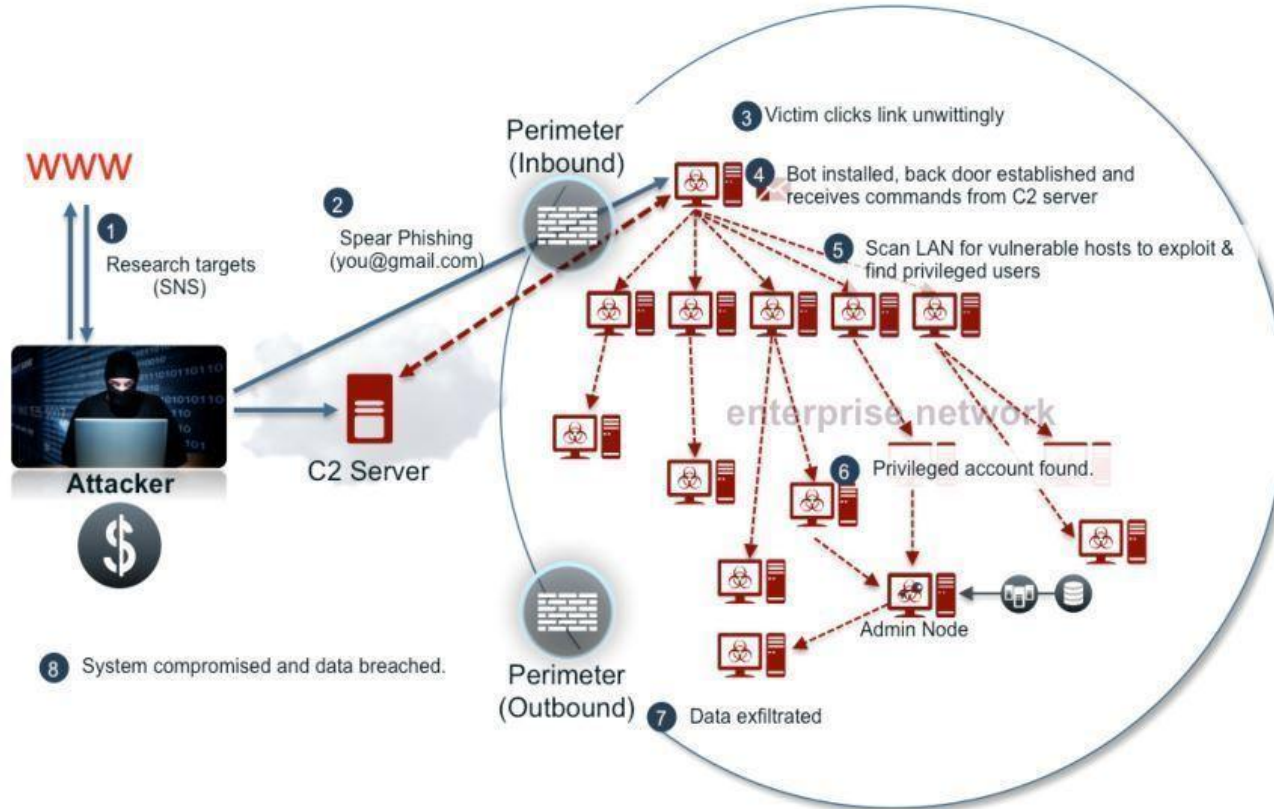
Attackers do this to:

- Find valuable data
- Reach domain controllers
- Access admin accounts
- Deploy ransomware across the network

Typical Attack Path

Phishing → User Laptop Compromise → Lateral Movement → Domain Controller → Full Network Control

Lateral Movement





Lateral Movement

Why Attackers Perform Lateral Movement

Because the **first compromised system** rarely has valuable data.

Example:

Attacker compromises:

Employee Laptop

But target is:

File Server

Domain Controller

Database Server

So attacker moves across the network.



Common Lateral Movement Techniques

Pass-the-Hash (PtH)

Attacker steals password hashes from memory.

Instead of cracking password, they **reuse the hash directly**.

Example tools:

- Mimikatz
- CrackMapExec

SOC Indicators:

- NTLM authentication anomalies
- Logins without password usage



Common Lateral Movement Techniques

Pass-the-Ticket (Kerberos Attack)

Attacker steals Kerberos authentication tickets.

Then uses them to access other machines.

SOC indicators:

- Suspicious Kerberos ticket usage
- Unusual TGT activity



Common Lateral Movement Techniques

Remote Desktop Protocol (RDP)

Attacker uses stolen credentials to connect via:

Port 3389

SOC indicators:

- Internal RDP connections
- Unusual login locations
- Login outside working hours



Common Lateral Movement Techniques

SMB / Windows Admin Shares

Attacker copies tools across machines using:

`\\target\C$`

Common tools:

- PsExec
- Impacket
- CrackMapExec

SOC indicators:

- Admin share access
- Suspicious file transfers
- Service creation events



Common Lateral Movement Techniques

Windows Management Instrumentation (WMI)

Allows remote command execution.

Example command:

```
wmic /node:target process call create "malware.exe"
```

SOC indicators:

- WMI process creation
- Remote execution logs



Real SOC Scenario

Alert sequence:

- Suspicious login to user laptop
- Mimikatz executed
- Admin credentials dumped
- Attacker logs into file server
- Ransomware deployed

Step 4 is **Lateral Movement**.

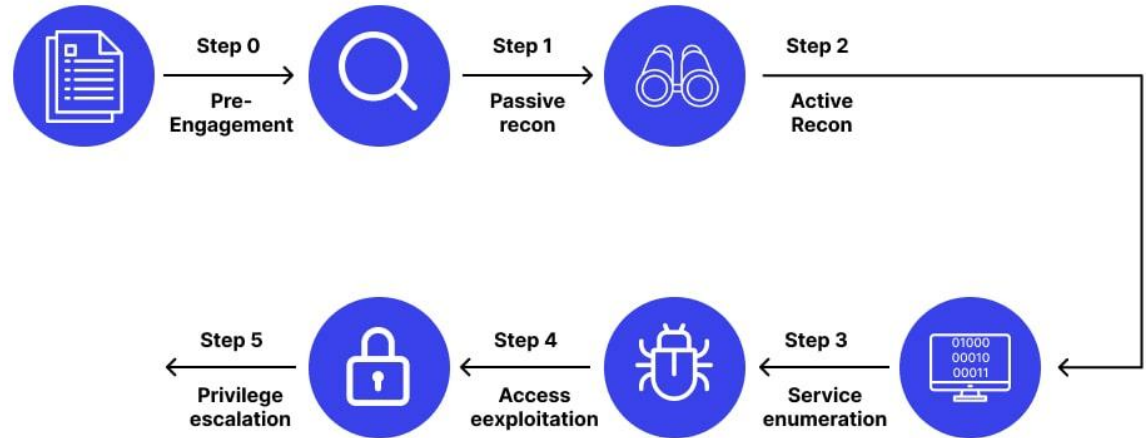


Privilege Escalation

Privilege escalation is when an attacker **gains higher permissions than originally allowed**.

Example:

User account → Administrator access





Privilege Escalation - Two Types

Vertical Privilege Escalation

User → Admin

Example:

Normal User → Local Admin → Domain Admin

Horizontal Privilege Escalation

User → Another User

Example:

Employee account → Manager account

Same level but different privileges.

Privilege Escalation

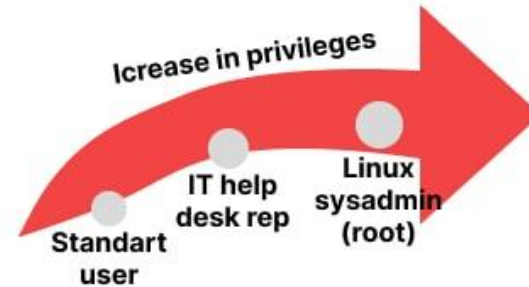


Horizontal privilege escalation attack



An internal horizontal escalation attack can occur between each of these accounts. Attackers can also achieve horizontal escalation by initially compromising one of these accounts.

Vertical privilege escalation attack



A threat actor moves from standard user access to an IT Help Desk account before gaining Linux sysadmin (root) access. Each subsequent step represents more privileged access

Why Attackers Need Privilege Escalation



Because initial compromise often gives:

Low privilege access

But attackers want:

Administrator / Root / Domain Admin

Which allows them to:

- Disable security tools
- Dump credentials
- Access servers
- Deploy ransomware



Exploiting Software Vulnerabilities

Example:

- Outdated Windows kernel
- Vulnerable drivers

Attacker runs exploit → becomes admin.

Credential Dumping

Tools like:

- Mimikatz
- LSASS memory dumping

Extract admin passwords.

SOC indicators:

- LSASS process access
- Credential dumping alerts



Misconfigured Permissions

Example:

Service runs as SYSTEM but executable is writable.

Attacker replaces file → gains admin access.

Token Impersonation

Windows access tokens stolen from high-privilege processes.

Attacker impersonates admin.

Scheduled Task Abuse

Attacker modifies scheduled tasks running as admin.



Watch for:

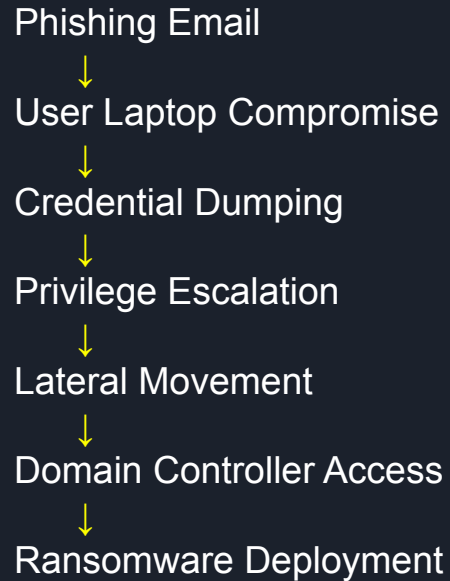
- Sudden admin rights assignment
- New admin account creation
- Service creation events
- Suspicious LSASS access
- Privilege escalation exploits

Important Windows Event IDs:

Event ID	Meaning
4672	Special privileges assigned
4688	Process creation
4720	User account created
4732	User added to admin group



Real ransomware scenario:





Difference Between Lateral Movement & Privilege Escalation

Feature	Lateral Movement	Privilege Escalation
Goal	Move across systems	Gain higher permissions
Direction	Sideways	Upwards
Example	Laptop → Server	User → Admin
Purpose	Expand access	Gain control



SOC L1 Thinking Task :

Suppose you see this sequence in logs:

1. User laptop executes Mimikatz
2. LSASS memory accessed
3. Admin login occurs from that machine
4. RDP connection to file server

Which stages occurred here?

Try to identify:

- Privilege Escalation
- Lateral Movement



Answer Of Task :

Step	Activity	Stage
1	Mimikatz executed	Credential Dumping
2	LSASS accessed	Credential Dumping
3	Admin login occurs	Privilege Escalation
4	RDP to file server	Lateral Movement

The stages that occurred are:

Credential Dumping → Privilege Escalation → Lateral Movement